YEAR 10 SCIENCE (EXTENSION)
PHYSICAL SCIENCES
EXAMINATION REVISION - SEMESTER 2 2019

1. A top sprinter reaches 11.2 ms^{-1} within 3.15 s of the start of a race.

   (a) Calculate the athlete's average acceleration.

   (b) How far does the sprinter cover while accelerating?

2. (a) How far does a car travel in a straight line if it accelerates at 1.4 ms^{-2} from 40 kmh^{-1} to 90 kmh^{-1}?

   (b) How long does this take?

3. How far does a Formula 1 race car travel as it accelerates at 8.70 ms^{-2} from 150 kmh^{-1} to 290 kmh^{-1} as it enters a straight?

4. A truck moving at 100 kmh^{-1} brakes uniformly at 1.5 ms^{-2}.

   (a) What is its final velocity after 5.0 s?

   (b) How far does it travel in this time?

5. A car of mass 1800 kg moving at 90 kmh^{-1} applies its brakes, exerting a force of 2300 N for 3.5 s. How far does it travel while braking?

6. A boy drops a stone down a disused well and hears it strike the bottom 2.40 s later.

(a) How deep is the well?

(b) What is the impact velocity of the stone?

7. Steven kicks a football vertically at 10 ms^{-1} in his backyard. It rises to a maximum height but gets caught in a tree on the way down, finishing 2.90 m above the point where it is kicked. Ignore any sideways motion of the ball.

(a) How high does the football rise?

(b) What is the ball's velocity when it gets caught by the tree?

(c) How long is the time of flight for the ball?

8. A car of mass 1530 kg is travelling at 70.0 kmh^{-1} along Hepburn Avenue when the driver has to brake sharply to avoid a small child who suddenly appeared on a bike crossing the road.

(a) If it took 3.50 s to stop the car, calculate the average force exerted by the brakes.

(b) The driver of the car was an engineer and had a hard-hat sitting on the back shelf. Describe what happened to the hard-hat in this situation and explain why it took place.

9. (a) Explain the difference between ***mass*** and ***weight***.

(b) "Space is great! You are weightless and it is easy to move stuff around." Comment on the accuracy of this statement.

(c) The acceleration due to gravity on Neptune is 11.1 ms^{-2}.

(i) Calculate the weight of a 200 kg object on Jupiter.

(ii) How long does it take for an object to fall from 4.0 m above Neptune's surface.

10. A 70.0 kg cyclist is applying a force of 150 N forwards but is working against a wind producing a friction force of 80 N opposing the movement.

(a) Determine the resultant force for this situation.

(b) Calculate the acceleration of the cyclist.

(c) It is possible for the rider to move at constant velocity. In terms of the forces acting, explain how this is possible.

11. You are sitting on a seat at the moment. Explain why you don't fall through the chair and are supported by it.

12. Steven enjoys BASE jumping off cliffs. He particularly enjoys a high cliff in Pakistan called Great Trango, which is 1340 m high.

    (a) What is Steven's theoretical velocity after falling for 14.0 s?

    (b) How far has Steven fallen during this time?

    (c) Why is your answer to part (a) theoretical?

    (d) Steven, who has a mass of 85 kg, is actually moving at 45.0 ms^{-1} at this point when he releases his parachute. What force does it exert on Steven if he decelerates to 25.0 ms^{-1} in 1.4 s?